# Seongwon (Gabriel) Yoon

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## Research Interests

Optical Interconnects, Silicon Photonics, Network Architecture, AI Infrastructure

## Education

**Georgia Institute of Technology** — Atlanta, GA, USA
*Ph.D. in Electrical & Computer Engineering (PI: Prof. Shimeng Yu)* — *Aug. 2024 – May. 2029 (expected)*

**Seoul National University** — Seoul, Korea
*B.S. in Materials Science and Engineering, Summa Cum Laude (GPA 4.11/4.3)* — *Mar. 2015 – Aug. 2022*

**Korea Science Academy of KAIST** — Busan, Korea
*High School Diploma* — *Mar. 2012 – Feb. 2015*

## Experience

**Georgia Institute of Technology** — Atlanta, GA, USA
*Graduate Research Assistant, Laboratory for Emerging Devices and Circuits* — *Aug. 2024 – Present*

- Co-designed collective communication (all-reduce, all-to-all) with photonic switch configuration and GPU placement to run LLM traffic at full link rate.
- Laid out a silicon photonic MZI switch (SiEPIC PDK) and simulated it in Ansys Lumerical FDTD, MODE, HEAT, and INTERCONNECT.
- Extended the htsim packet-level simulator with new GPU fabrics and calibrated NVSwitch baselines for MoE training and decode.
- Built a full-package Ansys thermal model of a co-packaged optical GPU node and its laser, ring-tuning, and energy budgets.

**Neuromorphic Materials and Devices Lab (NMDL)** — Seoul, Korea
*Undergraduate Researcher (PI: Prof. Sangbum Kim)* — *Mar. 2019 – Oct. 2021, Jun. 2023 – Aug. 2024*

- 1.4M 6T2R phase change memory synaptic arrays with stochastic leaky-integrate-and-fire neurons simulation

**Imagoworks Inc.** — Seoul, Korea
*Software Engineer, Deep CAD Team* — *Jul. 2022 – Feb. 2023*

- Fine-tuned generative models for 3D image dental solutions

**Samsung Electro-Mechanics** — Suwon, Korea
*Research Intern, Materials Research Group* — *Jan. 2022 – Feb. 2022*

- Analyzed TEM data for MLCC ceramic dielectric

**Bio-inspired Materials Lab** — Seoul, Korea
*Undergraduate Researcher (PI: Prof. Kitae Nam)* — *Dec. 2018 – Feb. 2019*

- Fabricated carbon nanotube embedded Mg-based hydrogen generating catalysts

## Publications

- **Seongwon Yoon**, Shimeng Yu, An Optical KV Cache Network for Disaggregated LLM Serving, arXiv preprint arXiv:2610.18444, 2026
- **Seongwon Yoon**, Pin-Jun Chen, Shimeng Yu, Cross-Layer Analysis of Thermal Tuning Stalls in Wafer-Scale Optical Interconnects for LLM MoE Training, arXiv preprint arXiv:2608.24637, 2026
- Faaiq Waqar, Ming-Yen Lee, **Seongwon Yoon**, Seongkwang Lim, Shimeng Yu, CMOS+X: Stacking Memories Based on Oxide Transistors upon GPGPU Platforms, International Symposium on Memory Systems (MEMSYS 2025)
- **Seongwon Yoon**, Janak Sharda, Omkar Phadke, Shimeng Yu, A CMOS Image Sensor Pixel Readout Circuit Design Using Back-End-of-Line (BEOL)-Compatible Oxide Transistors, IEEE International Midwest Symposium on Circuits and Systems (MWSCAS 2025), **Best Paper Award**

- Yu Gyeong Kang, Masatoshi Ishii, Jaeweon Park, Uicheol Shin, Suyeon Jang, **Seongwon Yoon**, Mingi Kim, Atsuya Okazaki, Megumi Ito, Akiyo Nomura, Kohji Hosokawa, Matthew BrightSky, Sangbum Kim, Solving Max-Cut Problem Using Spiking Boltzmann Machine Based on Neuromorphic Hardware with Phase Change Memory, Advanced Science, 2024
- Jungeun Park, **Seongwon Yoon**, Hannah Kim, Youngjun Kim, Uilyong Lee, Hyungseog Yu, Clinical Validity and Precision of Deep Learning-Based Cone-Beam Computed Tomography Automatic Landmarking Algorithm, Imaging Science in Dentistry, 2024
- **Seongwon Yoon**, Uicheol Shin, Sangbum Kim, Solving Constraint Satisfaction Problem with Spiking Neural Network Based on 1.4M 6T2R PCM Synaptic Array with 1.6K Stochastic LIF Neurons Neuromorphic Hardware, Korean Conference on Semiconductors (KCS 2021)

## Talks

**Optical Interconnect for LLM, MoE and Agentic AI** — Atlanta, GA, USA
*Future of CMOS and Talent Workshop, Memory Wall and Interconnect Wall, Georgia Tech* — *Mar. 2026*

## Technical Skills

**Photonic Simulation:** Ansys Lumerical (FDTD, MODE, HEAT, INTERCONNECT), Photonic PDK

**Device & Multiphysics:** TCAD, Ansys Mechanical (thermal and structural)

## Selected Honors and Awards

**Jung-hun Foundation Scholarship** — South Korea
*Full tuition support for 2 years* — *2018 – 2020*

**Eminence Scholarship, Seoul National University** — South Korea
*Full tuition support for 1.5 years* — *2015 – 2016*